higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

T130**(E)**(A9)T

NATIONAL CERTIFICATE

BUILDING AND STRUCTURAL CONSTRUCTION N4

(8060004)

9 April 2019 (X-Paper)
09:00–13:00

REQUIREMENTS: **ONE A2 drawing sheet**
Hot-rolled structural steel sections (red book) (BOE8/6)

This question paper consists of 5 pages.

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DEPARTMENT OF HIGHER EDUCATION AND TRAINING
REPUBLIC OF SOUTH AFRICA

NATIONAL CERTIFICATE
BUILDING AND STRUCTURAL CONSTRUCTION N4
TIME: 4 HOURS
MARKS: 100

INSTRUCTIONS AND INFORMATION

1. Answer ALL the questions.

2. Read ALL the questions carefully.

3. Number the answers according to the numbering system used in the question paper.

4. Strictly adhere to the following:

   Do QUESTIONS 1, 2 and 3 on the front of the DRAWING SHEET.
   Do QUESTIONS 5, 6 and 7 on the back of the DRAWING SHEET.
   Neatly print QUESTION 4 on the DRAWING SHEET.

5. Do NOT use an ANSWER BOOK.

6. The layout of answers is important.

7. Do ALL drawings in pencil with bold outlines.

8. Neatly print written answers in pencil.

9. Drawings must be done in accordance with National Standards and be fully labelled with descriptive notes and dimensions.

10. Work neatly.

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QUESTION 1

A single dwelling consists of cavity walls and a timber roof.

Draw, to scale 1:10, a vertical section through one of the eaves.

Use the following specifications:

Concrete foundation:	600 × 200 mm
Cavity wall:	270 mm wide The cavity below the DPC level must filled with weak concrete. The top FOUR courses, below the wall plate level, must be built solid. Show TWO tie wires in the wall. 19 mm plaster to the inside wall only
Natural ground level:	FOUR brick courses above the top of the foundation concrete
Roof construction:	300 mm overhang with 3 mm Flexit nailed to the underside of the rafters IBR roof sheeting on 75 × 50 mm purlins at 900 centres on 114 × 38 mm rafters, tie beam and wall plate 30° pitch
Ceiling:	6 mm Rhino board on 38 × 38 mm nailed brandering at 400 centres 75 mm Rhino cove cornice
Rainwater goods:	100 × 75 mm square gutter fixed to 230 × 32 fascia board nailed to the sprocket ends 75 mm diameter downpipe, shoe and wall brackets
Concrete floor:	THREE courses above the natural ground level finished off with 25 mm cement screed on 75 mm concrete on a damp-proof membrane on well-compacted clean filling

For drawing purposes use a floor-to-ceiling height of 1 250 mm. The completed drawing must include ALL labels, key symbols and a damp-proof course. **[22]**

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QUESTION 2

The span of a segmental gauged arch is 1 200 mm with a depth of 330 mm. The rise of the arch is 200 mm.

Draw, to scale 1:10, the front view of the arch. Clearly show the geometrical set-out on the left-hand side and face bricks on the right-hand side. Add ALL labels and dimensions to the finished drawing. **[12]**

QUESTION 3

Draw, to scale 1:5, a vertical section through a steel doorframe suitable for a one-and-a-half brick wide wall. The brick wall must be reinforced with THREE courses brick force and plastered on both sides. Show the position of the striking plate, rubber stops and finished floor level. **[11]**

QUESTION 4

4.1 Name each of the following types of flashing:

4.1.1 The strip of metal covering the apex where two sloping roofs meet

4.1.2 This covers the side wall and/or head-wall flashing

4.1.3 Where the higher end of the roof butts against a wall – the flashing is fixed to the roof and turned up and lifted free against the wall

(3 × 1) (3)

4.2 Centring is used where brick arches must be built.

Give a short description regarding the construction and purpose of centring for arches. (2)

4.3 Explain the concept of a *suspended ceiling*. (2)

[7]

QUESTION 5

A 254 mm × 254 mm × 72,9 kg/m H-section steel column is connected to a 500 × 500 × 30 mm base plate by means of TWO 15 mm thick gusset plates and TWO 150 × 90 × 10 mm rolled steel angles.

The column will be fillet welded to the base plate. The gusset plates and angles will be riveted using Θ16 snap-head rivets to the flanges of the steel column. The column will be anchored to an isolated concrete pad foundation by means of 4M12 holding-down bolts through FOUR Θ18 holes drilled 50 mm from the edges of the base plate.

Draw, to scale 1:10, the following views in first-angle orthographic projection:

5.1 A vertical section to show the column flange, gusset plate, angle, base plate and TWO holding-down bolts (9)

5.2 A top view of the column assembly (9)

Clearly show the layout of the snap-head rivets and 6 mm fillet welding symbols. The completed drawing must be fully labelled with the title and scale below each drawing. Do NOT show the concrete pad foundation. **[18]**

QUESTION 6

A light semicircular face-brick archway must be constructed over a 1 500 mm wide front door of a luxury dwelling. The construction for the centring consists of FOUR 32 mm thick ribs (TWO per side), TWO vertical central struts and TWO tie beams and TWO horizontal cross ribs. The 50 × 32 mm thick closed-lagging pieces are nailed across the top of the ribs. The timbers will be bolted together using M12 bolts. The centring will be supported on 150 × 50 mm bearers and steel telescopic metal struts secured to 230 × 38 mm sleepers.

Draw, to scale 1:10, the front view of the centring to show ALL components in place. The completed drawing must include ALL labels, dimensions and key symbols. Do NOT include the arch bricks. **[15]**

QUESTION 7

Draw, to scale 1:2, a horizontal section through a standard steel window built in the centre of a 270 mm wide cavity wall. The window is secured in the wall with window-fixing lugs. The external sill is finished off with a brick-on-edge sill while the inside sill consists of a 145 mm wide wooden board. The completed drawing must include the window frame, sash and glazing bar with glass and putty. Show the method of position, vertical damp-proof course and joint sealer. Include ALL labels, dimensions and wire ties in the cavity wall. **[15]**

TOTAL: 100

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